

L7a: Utility-Based Allocation and the Adaptive Rebalancing Engine

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Objectives for Today

Today we stop treating a portfolio as a vector of weights chosen once and treat it as a process: preferences computed every day from the single index model and the state of the market, answering which assets belong in the basket and how much of each, and a rebalancing engine with an account, rules, and a scorecard.

Today's concepts

- **Why a static allocation fails:** how buy-and-hold weights drift with prices and how to measure it, and the three ways the minimum-variance inputs let the optimizer down.
- **Utility-based allocation:** the budget-constrained problem, the Cobb–Douglas closed form, CES and its three limits, and preference weights from the SIM and two market inputs: signs choose the basket, magnitudes the weights.
- **The rebalancing engine:** a self-financing account with costs, next-bar execution, three trigger rules, and how to score a realized path.

Lectures and Examples

Lecture:

CHEME-5660-L7a-Lecture-Utility-Allocation-Rebalancing-Fall-2026.ipynb

Example 1:

CHEME-5660-L7a-Example-Utility-Allocator-Fall-2026.ipynb

Example 2:

CHEME-5660-L7a-Example-Adaptive-Rebalancing-Scorecard-Fall-2026.ipynb

Example 3:

CHEME-5660-L7a-Example-Portfolio-Drift-Fall-2026.ipynb

The first example builds the allocator and reads which firms belong in the basket; the second runs the engine through 2025 against buy-and-hold baselines; the third measures the drift of a tangent portfolio held through the year.

Concept Review: Allocation as a Time-Stamped Decision

L5b and L6b chose weights $\mathbf{w}$ over $|\mathcal{P}|$ risky assets from an expected growth-rate vector and covariance; with the SIM of L6a those inputs are $\hat{\mathbf{g}}_{\text{SIM}} = \hat{\alpha} + \hat{\beta} g'_M$ and $\hat{\Sigma}_{g,\text{SIM}} = s_{g,M}^2 \hat{\beta} \hat{\beta}^\top + \hat{\mathbf{D}}_g$ ($\hat{\mathbf{D}}_g$: residual variances).

- Target $g_\star$: $\mathbf{w}(g_\star)$ minimizes $\mathbf{w}^\top \hat{\Sigma}_g \mathbf{w}$ subject to $\mathbf{w}^\top \hat{\mathbf{g}} = g_\star$, $\mathbf{1}^\top \mathbf{w} = 1$, $w_i \geq 0$; the lowest point is $\mathbf{w}_{\text{GMV}}$; with a risk-free asset earning g_f , the fully invested point of the ray is the tangent portfolio $\mathbf{w}_{\mathcal{T}}$.
- Implemented on the decision date $t = 0$ by buying share counts, and held:

$$n_i = \frac{w_i W_{\mathcal{P}}(0)}{S_i(0)}, \quad W_{\mathcal{P}}(t) = \sum_{i \in \mathcal{P}} n_i S_i(t)$$

Two things to say out loud: the weights are optimal only for the inputs estimated on the decision date, and the investor's one decision was at $t = 0$. Today is about the other days.

Why Static Allocation Fails: Mechanical Drift

With fixed share counts (no cash, no dividends), $w_i(t) = n_i S_i(t) / W_{\mathcal{P}}(t)$; with $n_i = w_i(0) W_{\mathcal{P}}(0) / S_i(0)$ the initial wealth cancels:

$$w_i(t) = \frac{w_i(0) \overbrace{(S_i(t)/S_i(0))}^{\text{price relative}}}{\sum_{j \in \mathcal{P}} w_j(0) (S_j(t)/S_j(0))}$$

Weights stay put only if every held asset has the same price relative, which does not happen in practice. The drift on day t is the half- L^1 distance, the fraction of wealth a rebalance back to $w_i(0)$ would trade (before costs):

$$\frac{1}{2} \sum_{i \in \mathcal{P}} |w_i(t) - w_i(0)|$$

Example 3 (SIM tangent portfolio through 2025: drift 4.6% by year end):

CHEME-5660-L7a-Example-Portfolio-Drift-Fall-2026.ipynb

Why Static Allocation Fails: Fragile Inputs

Drift is the benign failure; the optimizer's inputs are the serious one. Three tendencies of estimated minimum-variance and tangent portfolios recur:

- **Input sensitivity**: estimated inputs are treated as exact, so estimation errors move the weights; the tangent portfolio overweights assets whose growth is overstated (Michaud's *error maximizer*, 1989), and errors in the means matter far more than errors in the variances (Chopra and Ziemba, 1993).
- **Concentration**: without binding constraints the tangent portfolio, and often the GMV portfolio, holds a few names (four of thirteen in L6b).
- **Assumption fragility**: one stable distribution is assumed, but correlations rise and volatilities jump in the episodes that matter.

Two responses: change the **objective**, writing preferences for each asset explicitly and letting them respond to the market state (today); or update the **inputs** as data arrive (L7b).

Utility Maximization

Instead of the smallest variance for a target, ask a consumer-choice question: given a budget and today's market, which portfolio does the investor prefer most? A utility function ranks portfolios, larger for more preferred. At time t , with prices $S_i(t)$ (USD per share) and wealth $W_{\mathcal{P}}(t)$ (USD), choose the share counts $n_i \geq 0$:

$$\begin{array}{ll} \underset{n_1, \dots, n_{|\mathcal{P}|}}{\text{maximize}} & U(n_1, \dots, n_{|\mathcal{P}|}) \\ \text{subject to} & \sum_{i \in \mathcal{P}} n_i S_i(t) = W_{\mathcal{P}}(t) \end{array}$$

- A **holding utility**: a preference over the shares held today at today's prices, not the expected utility of an uncertain terminal wealth; risk enters through the preferences and the engine's rules.
- **Two coupled questions**: which assets belong in the basket today, and how much of each. The utility functions answer the second; the preference model answers the first. Both are recomputed every trading day.

Cobb--Douglas Utility: Sign Classifies, Magnitude Allocates

Let $\gamma_i \in (-1, 1)$ be the preference weight of asset i . Its **sign** splits the assets into the preferred set $\mathcal{A}^+ = \{i : \gamma_i > 0\}$, the basket, and the non-preferred set $\mathcal{A}^- = \{i : \gamma_i \leq 0\}$, held at a floor of $n_{\min} \geq 0$ shares each; the utility is a product of powers over the preferred set, $U = \prod_{i \in \mathcal{A}^+} n_i^{\gamma_i}$ ($n_i > 0$ on $\mathcal{A}^+$, $n_i = n_{\min}$ on $\mathcal{A}^-$). With $W_{\text{adj}} = W_{\mathcal{P}}(t) - n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t) > 0$ the budget left after the floors and at least one preferred asset, the maximizer is closed form:

$$n_i^* = \underbrace{\frac{\gamma_i}{\sum_{j \in \mathcal{A}^+} \gamma_j}}_{\text{preference share}} \cdot \underbrace{\frac{W_{\text{adj}}}{S_i(t)}}_{\text{budget in shares}} \quad (i \in \mathcal{A}^+), \quad n_i^* = n_{\min} \quad (i \in \mathcal{A}^-)$$

The **dollar weight** of a preferred asset is its normalized preference weight $\gamma_i / \sum_{j \in \mathcal{A}^+} \gamma_j$, independent of prices. An empty basket means floors plus cash.

Cobb--Douglas Utility: Derivation

Maximizing U over $\mathcal{A}^+$ is maximizing $\ln U = \sum_{i \in \mathcal{A}^+} \gamma_i \ln n_i$ (the log is increasing). With multiplier λ on the budget over the preferred set, the first-order condition for asset i is:

$$\frac{\gamma_i}{n_i} = \lambda S_i(t) \quad \Longleftrightarrow \quad \underbrace{n_i S_i(t)}_{\text{dollars in } i} = \frac{\gamma_i}{\lambda}$$

Each preferred asset receives dollars in proportion to its exponent. Summing over $\mathcal{A}^+$ gives $W_{\text{adj}} = \sum_{j \in \mathcal{A}^+} \gamma_j / \lambda$, and substituting back gives the boxed result; strict concavity of $\ln U$ on the positive orthant makes it the unique maximum.

- The closed form applies to the preferred set only: a non-positive γ_i is a classification, not an exponent to insert (for $\gamma_i < 0$ the term $n_i^{\gamma_i}$ blows up as $n_i \downarrow 0$).
- The allocation responds immediately to a change in the preferences: no covariance matrix, no quadratic program, no numerical solver.

CES Utility: A Concentration Dial

For an elasticity of substitution $\eta > 0$, $\eta \neq 1$, the CES utility over the preferred set is $U_{\text{CES}}(\mathbf{n}) = (\sum_{i \in \mathcal{A}^+} \gamma_i n_i^{(\eta-1)/\eta})^{\eta/(\eta-1)}$ (γ_i now a coefficient). Its maximizer is closed form, share counts proportional to the bang for the buck raised to η :

$$n_i^* = W_{\text{adj}} \frac{(\gamma_i/S_i(t))^\eta}{\sum_{j \in \mathcal{A}^+} S_j(t)(\gamma_j/S_j(t))^\eta} \quad (i \in \mathcal{A}^+)$$

- $\eta \rightarrow 1$: the Cobb–Douglas allocation (its limit). $\eta \rightarrow \infty$: all of the preferred budget on the largest $\gamma_i/S_i(t)$ (when unique). $\eta \rightarrow 0^+$: equal share counts of every preferred asset.
- Engine’s CES variant: $\eta(\xi_t) = \eta_{\min} + (\eta_{\max} - \eta_{\min})/(1 + |\xi_t|)$, $0 < \eta_{\min} \leq \eta_{\max}$ (0.5, 5 in the examples): concentrate when the market-state signal ξ_t (an EMA crossover, two frames on) is quiet, diversify when it is loud. Proofs, log-linear utility, and a share-denomination caveat: advanced notebook.

Preference Weights from the Single Index Model

From the SIM parameters (α_i, β_i) and two market inputs on day t , the recent market growth rate $\tilde{g}_{M,t}$ and a market-state signal ξ_t :

$$\gamma_i(t) = \tanh \left(\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t} \tilde{g}_{M,t} \right)$$

The $\tanh$ bounds the weight in $(-1, 1)$. Its argument is the SIM's expected growth of asset i in today's market, $\alpha_i + \beta_i \tilde{g}_{M,t}$ (inverse years), through a **regime lens**: it equals $(\alpha_i + \beta_i \tilde{g}_{M,t}) / \beta_i^{\xi_t}$.

- **Which assets belong in the basket**: $\beta_i^{\xi_t} > 0$ and $\tanh$ keeps sign, so $\text{sign } \gamma_i(t) = \text{sign}(\alpha_i + \beta_i \tilde{g}_{M,t})$: preferred exactly when $\tilde{g}_{M,t} > -\alpha_i / \beta_i$. The signal never changes the classification; the recent market growth does.
- **How much of each**: the lens moves the magnitudes: bearish ($\xi_t > 0$) tilts toward low-beta names, bullish ($\xi_t < 0$) toward high-beta.

Preference Weights: The Basket, and Assumptions Stated

- As $\tilde{g}_{M,t}$ moves, assets **cross between the sets**: a strong market pulls in firms with a small negative α_i and a large β_i ; a weak market pushes out everything but the strongest intercepts.
- If every $\gamma_i(t) \leq 0$, the model is saying **this basket does not belong in this market**: the allocator holds the floors plus cash, and an investor not content with cash should rethink which assets are in the basket (the ticker-picking methods of L12b and L13a).
- **Assumptions**: (i) $\beta_i > 0$ (the course firms: 0.54 to 1.75, asserted; a non-positive beta needs a rule, e.g., assign to $\mathcal{A}^-$); (ii) the argument is an annualized growth rate read as a pure number at the one-year horizon; (iii) $\tilde{g}_{M,t}$ swings by about a unit per year, an order of magnitude more than the intercepts: that is what makes the basket respond.

The Market Inputs

Both come from SPY with exponential moving averages, $\bar{S}_t = \omega S_t + (1 - \omega)\bar{S}_{t-1}$, $\omega = 2/(L + 1)$, window L days. With $L_{\text{short}} = 21$, $L_{\text{long}} = 63$, and a dimensionless gain $G > 0$, the signal is the scaled crossover:

$$\xi_t = -G \left(\frac{\bar{S}_t^{\text{short}}}{\bar{S}_t^{\text{long}}} - 1 \right)$$

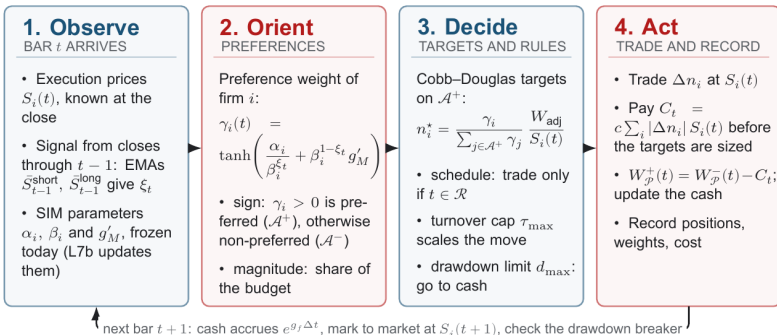
- Falling market (short EMA below long): $\xi_t > 0$, bearish; rising: $\xi_t < 0$, bullish; G sets $|\xi_t| \leq 1$ on 99% of training days ($G \approx 21$).
- $\tilde{g}_{M,t}$: the EMA (window $L_{\text{growth}} = 10$) of the daily market growth rates $g_{M,t} = \ln(S_t/S_{t-1})/\Delta t$. Both inputs on day t use prices through day $t - 1$.

Example 1: CHEME-5660-L7a-Example-Utility-Allocator-Fall-2026.ipynb (first trading day of 2025: $\tilde{g}_M = -0.76$ per year, no firm preferred)

The Rebalancing Engine

The rebalancing engine: one pass per trading day

Signal from yesterday, trade at today's price, pay the cost, record, repeat



A static allocation (L5b, L6b) is one pass on the decision date, then hold.

Every trading day is one pass: observe, orient, decide, act. A static allocation is one pass on the decision date, then nothing.

The Rebalancing Engine: The Account and the Timing

Let $n_{i,t}$ be the shares held after the close of day t ; cash earns g_f . At the close of day t , before any trade, the book is worth:

$$W_{\mathcal{P}}^{-}(t) = \underbrace{\text{cash}_{t-1} e^{g_f \Delta t}}_{\text{cash}} + \underbrace{\sum_i n_{i,t-1} S_i(t)}_{\text{marked to market}}$$

Trading share deltas Δn_i at the close $S_i(t)$ moves gross notional $Q_t = \sum_i |\Delta n_i| S_i(t)$ and costs $C_t = c Q_t$ (c the cost rate). The cost comes out of the book and the positions are sized against what is left, the two solved together:

$$\boxed{W_{\mathcal{P}}^{+}(t) = W_{\mathcal{P}}^{-}(t) - C_t}$$

so $\text{cash}_t + \sum_i n_{i,t} S_i(t) = W_{\mathcal{P}}^{+}(t)$ exactly and cash never goes negative. **Next-bar execution**: inputs and preferences for day t use prices through $t - 1$; the trade executes at day t 's close. A close-based signal traded at that close is look-ahead.

The Rebalancing Engine: Three Trigger Rules

Three rules bound the loop (else it trades daily, ignores a crash, or acts on every wiggle of the inputs):

- **Reallocation schedule** $\mathcal{R}$: trade only on days $t \in \mathcal{R}$ (daily, or every 21 days); hold otherwise.
- **Turnover cap** $\tau_{\max}$: with Δw_i the change of asset i 's weight and $\Delta w_{\text{cash}} = -\sum_i \Delta w_i$, the one-way turnover $\tau_t = \frac{1}{2}(\sum_i |\Delta w_i| + |\Delta w_{\text{cash}}|)$ is the fraction of wealth that changes hands one way; a move with $\tau_t > \tau_{\max}$ is scaled down (the cost applies to the gross Q_t).
- **Drawdown limit** $d_{\max}$: a circuit breaker checked every day before the schedule; if $1 - W_{\mathcal{P}}^-(t) / \max_{s \leq t} W_{\mathcal{P}}^-(s) > d_{\max}$, liquidate to cash (bypassing the cap), stay in cash n_{lock} more days, rebase the peak on the first scheduled trade after. It does not cap the recorded drawdown at $d_{\max}$.

They interact: all-cash to fully invested is a one-way turnover of one, so with $\tau_{\max} = 0.5$ the engine is only half invested until its next scheduled day.

Algorithm: Rebalancing Engine (Backtest), Setup

The engine is **allocator-agnostic**: it takes the prices and a target function, and our target functions wrap the allocators of this lecture around the lagged preference weights.

- **Given:** close prices $S_i(t)$, $t = 1, \dots, N$; the initial wealth $W_{\mathcal{P}}(0)$ in cash; the set $\mathcal{R}$; the rule parameters $(c, \tau_{\max}, d_{\max}, n_{\text{lock}})$.
- **Target function** $\mathbf{w}^*(t, \text{state})$: returns risky weights ($w_i^* \geq 0$, $\sum_i w_i^* \leq 1$, remainder cash) from the day and the current state (pre-trade wealth, weights, shares, cash, and a copy of the day's execution prices).
- **Ours:** compute $\gamma_i(t)$ from the market inputs through $t - 1$, allocate by Cobb–Douglas (or CES with $\eta(\xi_t)$) at the day's prices, return the allocation as weights; on an empty basket, floors plus cash.
- **Static baselines:** the same loop with a constant target and $\mathcal{R} = \{1\}$: same costs, same accounting.

Algorithm: Rebalancing Engine (Backtest), Daily Loop

For $t = 1, \dots, N$ (one bar at a time, under the timing rule):

1. Accrue cash by $e^{g_f \Delta t}$; mark to market at $S_i(t)$ to get $W_{\mathcal{P}}^-(t)$; update the running peak.
2. Breaker: if the drawdown exceeds $d_{\max}$ with risky positions held, liquidate at $S_i(t)$, pay c on the notional sold, lock n_{lock} more days, go to 5.
3. If $t \in \mathcal{R}$ and not locked: ask for $\mathbf{w}^*$; form the proposed change of weights, scale to $\tau_{\max}$.
4. Solve cost and positions together on the post-cost budget; execute Δn_i at $S_i(t)$; pay $C_t = c Q_t$; update the cash.
5. Record wealth, cash, positions, weights, turnover, cost, and flags.

Example 2: CHEME-5660-L7a-Example-Adaptive-Rebalancing-Scorecard-Fall-2026.ipynb

Scoring a Realized Path

Did the portfolio beat what the same dollars earn at the risk-free growth rate? For one outflow, one terminal value, constant g_f , horizon $T = N\Delta t$ years:

$$\text{NPV}(g_f, T) = \underbrace{-W_{\mathcal{P}}(0)}_{\text{investment}} + \underbrace{W_{\mathcal{P}}(T) e^{-g_f T}}_{\text{discounted terminal wealth}}$$

positive means the run beat cash at g_f ; the risk-free portfolio is the zero.

- **Realized growth** $\ln(W_{\mathcal{P}}(T)/W_{\mathcal{P}}(0))/T$; **maximum drawdown** $\max_t (1 - W_{\mathcal{P}}(t) / \max_{s \leq t} W_{\mathcal{P}}(s))$.
- **Realized Sharpe ratio**: mean of the daily excess log growth over its standard deviation, times $\sqrt{1/\Delta t}$ (annualized, as in L6b).
- **Turnover, cost, interventions**: $\sum_t \tau_t$, $\sum_t C_t$, breaker firings.

One row per run, these columns. **One path is one draw**: what each policy did, not how likely it was; fail rates and tail quantiles need many paths (L7b).

Scoring a Realized Path: 2025 in One Table

Example 2: $\mathcal{R}$ every 21 days, $\tau_{\max} = 0.5$, $d_{\max} = 0.10$, $c = 5$ basis points, one engine for every run. The model emptied the basket on 57 of 250 days and preferred all thirteen firms on 152.

- **Cobb–Douglas, monthly**: 12 scheduled trades, floors plus cash on 50 days, no breaker firing; ended at 1.53 times initial wealth, drawdown 9.0% (rules off: 1.52).
- **Cobb–Douglas, daily**: 228 trades, one breaker firing, twenty times the cost; 1.40 (10.8%). **CES with $\eta(\xi_t)$, monthly**: 1.29 (9.2%).
- **Buy and hold**: tangent 1.28 (drawdown 31%), GMV 1.38 (13%), equal weights 1.59 (24%), SPY 1.17 (19%).

Read across, then against the baselines: who beat cash, what the basket selection did on the bad days, what a schedule cost. One year, one draw.

Example 2: CHEME-5660-L7a-Example-Adaptive-Rebalancing-Scorecard-Fall-2026.ipynb

Optional Advanced Material

Advanced: [advanced/adaptive_utility/CHEME-5660-L7a-Advanced-CES-Limits-Fall-2026.ipynb](#)

Optional, not a prerequisite for L7b. Derives the CES closed form and proves its three limits (Cobb–Douglas at $\eta \rightarrow 1$, concentration at $\eta \rightarrow \infty$, equal share counts at $\eta \rightarrow 0^+$), reads the two limits of the elasticity rule $\eta(\xi_t)$, shows that log-linear utility has the Cobb–Douglas optimizer, and shows why the CES allocation over share counts depends on the price level of each share when $\eta \neq 1$.

Notebook: [lectures/week-7/L7a/advanced](#), also linked from the lecture notebook.

Summary

Today: drift and fragile inputs, the Cobb–Douglas and CES allocators, preferences from the SIM and two market inputs, and an engine with an account, rules, and a scorecard.

Key takeaways

- **Utility makes preferences explicit, Cobb–Douglas turns them into weights in closed form:** sign puts an asset in or out of the basket, magnitude sets its share of the preferred budget; CES is a concentration dial.
- **The SIM plus two market inputs supplies the preferences, the rules bound the response:** recent market growth chooses the basket (empty means rethink it, or hold cash), the signal tilts the magnitudes; the rules decide when, how far, and whether to act.
- **A realized-path scorecard is one draw:** NPV against cash, drawdown, growth, Sharpe, turnover, cost say what happened; the distribution needs many paths.

Next, L7b: SIM parameters that update as data arrive, and an ensemble of futures.

Today: preferences computed every day, an allocation in closed form, and an engine that pays its own costs.